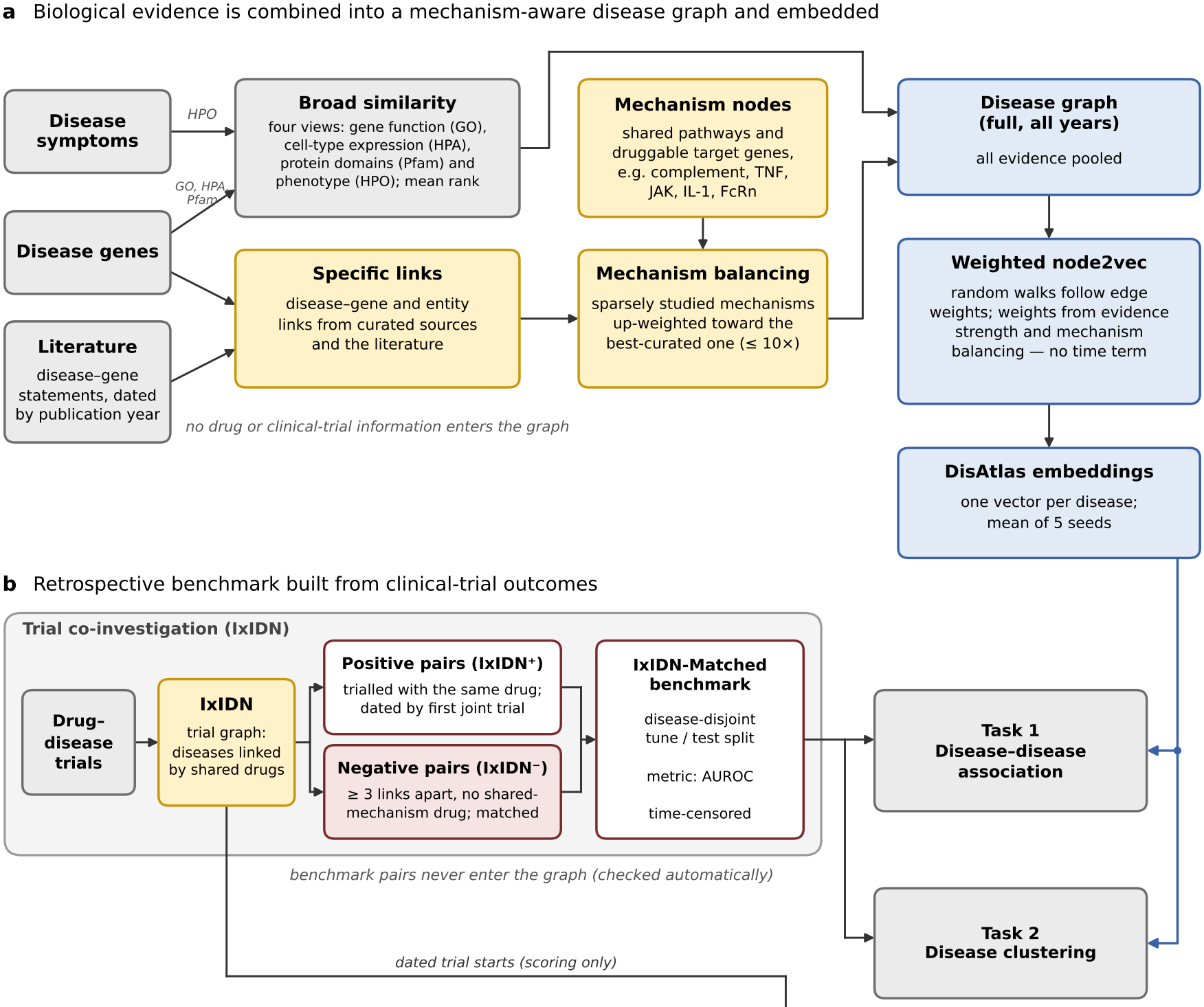

a Biological evidence is combined into a mechanism-aware disease graph and embedded
Disease symptoms
HPO
Broad similarity
four views: gene function (GO), cell-type expression (HPA), protein domains (Pfam) and phenotype (HPO); mean rank
Mechanism nodes
shared pathways and druggable target genes, e.g. complement, TNF, JAK, IL-1, FcRn
Disease graph (full, all years)
all evidence pooled
GO, HPA Pfam
Disease genes
Specific links
disease–gene and entity links from curated sources and the literature
Mechanism balancing
sparsely studied mechanisms up-weighted toward the best-curated one (≤ 10×)
Weighted node2vec
random walks follow edge weights; weights from evidence strength and mechanism balancing — no time term
Literature
disease–gene statements, dated by publication year
no drug or clinical-trial information enters the graph
DisAtlas embeddings
one vector per disease; mean of 5 seeds
b Retrospective benchmark built from clinical-trial outcomes
Trial co-investigation (IxIDN)
Drug–disease trials
IxIDN
trial graph: diseases linked by shared drugs
Positive pairs (IxIDN⁺)
trialled with the same drug; dated by first joint trial
Negative pairs (IxIDN⁻)
≥ 3 links apart, no shared-mechanism drug; matched
IxIDN-Matched benchmark
disease-disjoint tune / test split
metric: AUROC
time-censored
Task 1 Disease–disease association
Task 2 Disease clustering
benchmark pairs never enter the graph (checked automatically)
dated trial starts (scoring only)
c Simulated prospective test, year by year
Dated edges
each graph edge in (a) dated by its first evidence
Yearly graph snapshots
evidence up to the end of year y−1
Snapshot embeddings
DisAtlas retrained on each snapshot
Score each drug
candidates: diseases not yet trialled with the drug; score = highest similarity to the drug's earlier trial diseases
Rank its new trials
percentile of each disease the drug first enters in year y, among diseases of similar trial activity (and degree), vs weighted Adamic–Adar